

The Computer Networks

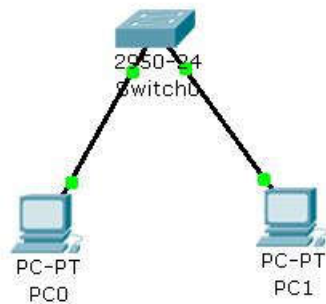
Lab 2

1. Creating a simple switched network

The goal of this first part is to create and test a switched network between two PCs.

1.1 Implementation of the model

Realizing the diagram below:



1. Which cable you need to use to connect the two PCs to the switch?
2. What color has the port of the switch when connecting the PC?
3. What happens at that time?
4. What color is the switch port sometime after?

1.2 Assigning an address to PCs and tests

You will now assign the two PCs the following IP addresses:

Equipment	IP address
PC0	192.168.1.10 / 24
PC1	192.168.1.11 / 24

1. For the configuration of PCs, you do not need to enter a gateway address. Why ?
2. Use the command "ipconfig / all" to verify with certainty the IP address of PCs. This command must be launched from the command prompt.
3. Test the connection between the two PCs using a ping and make sure the connection is working.

4. Launch a frame analysis, retest the link, and complete the following table by looking inside the PDU of the ICMP:

ICMP	Code	Type
Echo (Ping) Request		
Echo (ping) reply		

You will now change the address of PC1 with the following IP address:

Equipment	IP address
PC1	10.1.0.10 / 16

1. Testing the connection between the two PCs through a ping.
2. Does the link work?
3. In justifying your answer, what **equipment** should have been added to make it work?

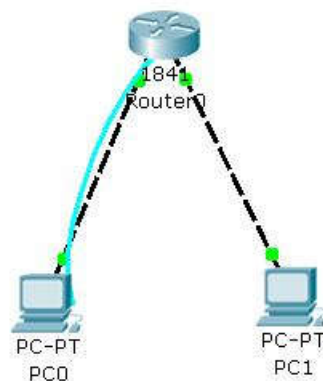
2. Simple routed network

The purpose of this second part is to study a simple routed network to correct the problem encountered last.

WARNING!!! For now, do not change the configuration of your PCs.

2.1 Implementation of the model

The model set up is as follows:



With

Equipment	IP address
Fa0 / 0	192.168.1.1 / 24
Fa0 / 1	10.1.0.1 / 16

We will now configure the router (after switching on of course) so it becomes a gateway. For this, from the PC connected to the router via the console cable, run the hyperterminal application (Desktop -> Terminal).

Normally, you should see the router starts, and then ask yourself this question:

Continue with Setup Dialog [yes / no]?

To which you respond with a No that you will follow several input (the button _)

What is the prompt that appears now?

Become router administrator by typing enable.

What is the prompt that appears?

Using the “configure terminal” command, enter configuration mode.

What is the prompt that appears?

Now enter the following commands:

```
Router(config) #fa0/0 interface
```

```
Router(config-if) #ip address 192.168.1.1 255.255.255.0
```

```
Router(config-if) #no shutdown
```

```
Router(config-if) #exit
```

```
Router(config) #fa0/1 interface
```

```

Router(config-if) #ip address 10.1.0.1 255.255.0.0
Router(config-if) #no shutdown
Router(config-if) #exit
Router(config-if) #exit
Router #show ip int brief

```

What is the state of interfaces FastEthernet0/0 and FastEthernet0/1?

2.2 Tests

Indicate in the table if the following tests are not working

Test	Result
Ping between PC0 and fa0/0	
Ping between PC0 and fa0/1	
Ping between PC1 and fa0/1	
Ping between PC1 and fa0/0	
Ping between PC1 and PC0	

Change the IP configuration of your PC by the address of its gateway.

Repeat the previous tests and complete the following table:

Test	Result
Ping between PC0 and fa0/0	
Ping between PC0 and fa0/1	
Ping between PC1 and fa0/1	
Ping between PC1 and fa0/0	
Ping between PC1 and PC0	

Finally, fill in the following table:

	Package from Pc0	Package arriving to Pc1
IP Address Source		
Destination IP Address		
Source Mac Address		
Destination Mac Address		